

57

A nine-year-old male child was brought by his parents to a surgical emergency OPD with complaint of abdominal pain and anorexia of four days duration. The pain was initially periumbilical then it involves the whole abdomen. The patient also had high-grade intermittent fever. On physical examination, he was acutely sick looking, pulse rate was 128/min and respiratory rate was 82/min and temperature was 99.6 degree centigrade. The abdomen was tender and rigid. What is the most likely diagnosis?

- ☐ Simple appendicitis
- ☐ Mesenteric lymphadenitis
- ☐ Perforated PUD
- ☒ Perforated appendicitis

Explanation :

The child has clear signs of advanced intra-abdominal infection with peritonitis. The clinical presentation of appendicitis with delayed diagnosis fits the timeline for perforation (typically after 48 hours).

- Schwartz's Principles of Surgery: Management of appendicitis.

58

A 62-year-old male patient, who was stabbed by the horn of an Ox on to his back, came to a clinic with a complaint of difficulty of moving his left leg. His bowel and bladder control is intact. On physical examination, vital signs are in normal range and on nervous system examination, the tone of left leg is increased, the power is zero on the left side and 5/5 on the right leg and sensation is lost on the right leg. What is the most likely diagnosis in this case?

- ☒ Brown-Sequard syndrome
- ☐ Anterior Cord syndrome
- ☐ Cauda Equina syndrome
- ☐ Central cord syndrome

Explanation :

The combination of ipsilateral motor and proprioceptive loss with contralateral sensory loss strongly suggests Brown-Sequard syndrome (BSS). The mechanism of trauma (ox horn stab to the back) aligns with spinal cord hemisection.

- Bradley's Neurology, UpToDate: "Clinical presentation and diagnosis of traumatic spinal cord injuries."

59

A 5-year-old boy is brought to the ED by his parents because of multiple episodes of bruising and petechiae over the past few days. The child appears well and has no significant past medical history. On examination, the child has scattered petechiae and ecchymoses, but no hepatosplenomegaly or lymphadenopathy. CBC reveals a platelet count of 6,000/ μ L. The rest of the CBC is within normal limits, and other cell lines are not affected. The clinical presentation and lab results suggest immune thrombocytopenia. What is the first step in management?

- ☐ Administer corticosteroids
- ☐ Transfuse platelets
- ☒ Administer IVIG

A 5-year-old boy is brought to the ED by his parents because of multiple episodes of bruising and petechiae over the past few days. The child appears well and has no significant past medical history. On examination, the child has scattered petechiae and ecchymoses, but no hepatosplenomegaly or lymphadenopathy. CBC reveals a platelet count of $6,000/\mu\text{L}$. The rest of the CBC is within normal limits, and other cell lines are not affected. The clinical presentation and lab results suggest immune thrombocytopenia. What is the first step in management?

☐ Administer corticosteroids

☐ Transfuse platelets

☒ Administer IVIG

☐ Splenectomy

Explanation :

Immune thrombocytopenia, also known as idiopathic thrombocytopenic purpura or immune thrombocytopenic purpura is the most common cause of acute onset of thrombocytopenia in an otherwise well child. The exact antigenic target for most such antibodies in most cases of childhood acute ITP remains undetermined, although in chronic ITP many patients demonstrate antibodies against $\text{aIIb-}\beta 3$ and GPIb. After binding of the antibody to the platelet surface, circulating antibody-coated platelets are recognized by the Fc receptor on splenic macrophages, ingested, and destroyed. The classic presentation of ITP is a previously healthy 1-4 yr old child who has sudden onset of generalized petechiae and purpura. The parents often state that the child was fine yesterday and now is covered with bruises and purple dots. There may be bleeding from the gums and mucous membranes, particularly with profound thrombocytopenia (platelet $<10\text{k}$). As American Society of Hematology guidelines state, "A single dose of IVIG (0.8-1.0 g/kg) or a short course of corticosteroids should be used as first-line treatment." IVIG for 1-2 days induces a rapid rise in platelet count (usually $>20\text{k}$) in 95% of patients within 48 hr, providing a temporary boost in platelet numbers while waiting for other treatments, such as corticosteroids, to take effect.

Corticosteroid therapy has been used for many years to treat acute and chronic ITP in adults and children. Doses of prednisone at 1-4 mg/kg/24 hr appear to induce a more rapid rise in platelet count than in untreated patients with ITP. Corticosteroid therapy is usually continued for short course until a rise in platelet count to $>20\text{k}$ has been achieved to avoid the long-term side effects of corticosteroid therapy, especially growth failure, diabetes mellitus, and osteoporosis. IVIG is often preferred for rapid elevation of platelet counts.

Platelet transfusions are generally reserved for life-threatening bleeding in ITP because the transfused platelets are quickly destroyed by the immune system.

The role of splenectomy in ITP should be reserved for 2 circumstances: (1) Age $\geq 4\text{yr}$ with severe ITP that has lasted $>1\text{yr}$ (chronic ITP) and whose symptoms are not easily controlled with therapy and (2) when life-threatening hemorrhage (ICH) complicates acute ITP, if the platelet count cannot be corrected rapidly with transfusion of platelets and administration of IVIG and corticosteroids.
- Nelson 21st p4608-4610, UpToDate

60

A 24-year-old woman in her first trimester of pregnancy presents to the clinic with a two-week history of severe nausea and vomiting. She reports being unable to keep food or liquids down and has lost 5% of her pre-pregnancy body weight. She denies abdominal pain, fever, or diarrhea. On examination, she appears dehydrated, and her urine dipstick test shows ketones +3.

☐ Esophageal rupture

☒ Wernicke's encephalopathy

☐ Acute tubular necrosis



A 24-year-old woman in her first trimester of pregnancy presents to the clinic with a two-week history of severe nausea and vomiting. She reports being unable to keep food or liquids down and has lost 5% of her pre-pregnancy body weight. She denies abdominal pain, fever, or diarrhea. On examination, she appears dehydrated, and her urine dipstick test shows ketones +3.

- ☐ Esophageal rupture
- ☒ Wernicke's encephalopathy
- ☐ Acute tubular necrosis
- ☐ Preeclampsia

Explanation :

Serious Maternal life threatening complications of HEG can include Wernicke's encephalopathy, acute tubular necrosis, central pontine myelinolysis, Mallory-Weiss tear of the esophagus, pneumomediastinum, and splenic avulsion. The most feared complication of hyperemesis gravidarum (severe nausea and vomiting in pregnancy) is Wernicke's encephalopathy due to thiamine (vitamin B1) deficiency. It can lead to permanent brain damage if not promptly treated with thiamine supplementation.

Esophageal rupture severe vomiting can lead to esophageal rupture (Boerhaave syndrome), it is less common than other complications and typically presents with chest pain and signs of mediastinitis.

Dehydration can lead to prerenal azotemia and acute kidney injury, but it is not the most feared complication as it is often reversible with fluid resuscitation.

- William's Obstetrics 26th p2611-2612, Current ObGyn 11th p911

A 30-year-old male patient came to an OPD with complaint of productive cough, pleuritic chest pain and fever of six days duration. On examination, his blood pressure is 90/60mmHg respiratory rate is 28/min, temperature is 38.4 degree centigrade, and there is dullness and absent air entry on the left lower third of chest. Chest x-ray reveals left pleural effusion. Pleural fluid analysis shows LDH level of 800, glucose level of 80mg/dl, WBC count 15000/mm³, and gram-positive cocci in chains on gram stain. What is the most likely indication for insertion of chest tube in this case?

- ☐ Pleural LDH level
- ☒ Pleural gram stain result
- ☐ Pleural WBC count
- ☐ Pleural glucose level

Explanation :

Empyema diagnosis or Positive Gram stain (gram-positive cocci in chains) confirms infection in pleural space which is an indication for Chest Tube Insertion. Immediate chest tube insertion is necessary to drain infected fluid, prevent loculation, and allow for resolution of infection. Without drainage, empyema can lead to sepsis or fibrothorax.

- Harrison's 21st, UpToDate: "Management of complicated parapneumonic effusions and empyema in adults."

A significant number of prisoners visited a nearby hospital due to epidemic typhus and louse borne



62

A significant number of prisoners visited a nearby hospital due to epidemic typhus and louse borne relapsing fever. The manager of the prison consulted health workers or the hospital to suggest sustainable solution besides diagnosis and treatment. What is the most appropriate control mechanism of the disease in this case?

☐ Cleaning the surrounding

☐ Boiling cloths

☒ Insecticide

☐ Personal hygiene

Explanation :

Control Mechanisms are:

- Insecticide application (e.g., permethrin, malathion) is the most effective intervention for eliminating lice in large populations.
- Mass delousing with insecticide offers immediate and sustainable impact, reducing the transmission of louse-borne diseases. Directly targets the vector (lice) and prevents re-infestation. Practical for large-scale application in settings like prisons.
- CDC Guidelines on Louse-Borne Diseases.

63

A manager of a Zonal Health Department involves her subordinates in decision making by consulting team members. She values group discussions and input from team members: she maximizes the members' strong points in order to obtain the best performance from the entire team. What is the most likely leadership style the manager applied to carry out her roles?

☐ Free-rein

☐ Laissez-faire

☐ Autocratic

☒ Participative

Explanation :

Participative (Democratic) leadership style involves collaboration and input from team members in decision-making. It values communication, teamwork, and the strengths of individual members to optimize overall performance. The manager fosters a sense of shared responsibility, ensuring that decisions are well-informed and broadly supported. Consulting team members and encouraging group discussions are hallmark features of participative leadership.

Free-rein implies minimal managerial intervention and allows employees to make decisions independently, which is not described here. Laissez-faire a hands-off style that delegates decision-making entirely to subordinates, with limited consultation or group discussions. Autocratic involves unilateral decision-making by the leader without consulting team members, contrasting with the consultative approach described.

64

A 45-year-old male patient came to an emergency department with a complaint of abdominal pain and coffee ground vomiting of three days duration. He used to take Prednisolone from nearby pharmacy for asthma without any prescription. On evaluation, his pulse rate was 108/min and blood pressure was 90/60mmHg. He had diffused abdominal tenderness and rigidity. What is the most effective initial management for this patient?



A 45-year-old male patient came to an emergency department with a complaint of abdominal pain and coffee ground vomiting of three days duration. He used to take Prednisolone from nearby pharmacy for asthma without any prescription. On evaluation, his pulse rate was 108/min and blood pressure was 90/60mmhg. He had diffused abdominal tenderness-and rigidity. What is the most effective initial management for this patient?

- ☐ Send him for erect chest x-ray
- ☒ Fluid resuscitation, antibiotics and serum amylase

☐ Fluid resuscitation and antibiotics

☐ Fluid resuscitation, antibiotics and CT-scan

Explanation :

This presentation is suggestive of perforated peptic ulcer with peritonitis. Risk factors include NSAIDs and corticosteroid use, which impair gastric mucosal defenses. Coffee-ground vomiting indicates upper gastrointestinal bleeding. Initial Management starts with fluid Resuscitation, Antibiotics; and determine serum Amylase.

- Sabiston Textbook of Surgery, UpToDate: "Evaluation and management of acute perforated peptic ulcer disease."

A mother claimed that her 24-hour-old neonate does not urinate. On examination, the neonate's bladder is palpable and has well-developed external genitalia. Ultrasound shows thickened bladder wall with bilateral hydronephrosis. There is no evidence of vesicoureteral reflux on voiding cystourethrogram. What is the most likely diagnosis?

☐ Vesicoureteral reflux

☒ Posterior urethral valve

☐ PUJ obstruction

☐ Multicystic dysplastic kidney

Explanation :

PUV is a congenital obstruction of the posterior urethra caused by abnormal membranous folds, leading to bladder outlet obstruction. Palpable bladder and hydronephrosis strongly suggest bladder outlet obstruction.

PUV is the most common etiology in neonates:

- Nelson 21th, UpToDate: "Congenital causes of urinary tract obstruction in neonates."

A health center has planned to provide long-acting modern family planning services for 500 users in its annual operational plan. At the end of the year, the center assesses its performance and found that 510 women utilized the services. What is the most appropriate conclusion regarding the plan and the performance?

☒ Effective and Efficient

☐ Efficient and Equitable

☐ Effective

☐ Efficient

Explanation :



A health center has planned to provide long-acting modern family planning services for 500 users in its annual operational plan. At the end of the year, the center assesses its performance and found that 510 women utilized the services. What is the most appropriate conclusion regarding the plan and the performance?

☒ Effective and Efficient

☐ Efficient and Equitable

☐ Effective

☐ Efficient

Explanation :

The goal of the annual operational plan was to serve 500 users, and the center exceeded this target by providing services to 510 women. This indicates the plan was effective because it achieved its intended outcome. Efficiency measures how well resources (time, budget, personnel) are utilized to achieve the target. By exceeding the goal, the center demonstrated efficiency in utilizing resources effectively to provide services.

Equity isn't explicitly assessed in the scenario since there is no mention of access disparities or inclusivity.

- WHO Performance Standards: "Effectiveness and efficiency in health system performance."

A manager of a Zonal Health Department annually gives orientation to the heads of district health offices on the main components of the 1993 health policy of the Transitional Government of Ethiopia and how to incorporate this policy areas to their strategic plan. What component of this health policy could be the focus area for the heads?

☐ Give a special focus on the construction of new specialize hospitals

☐ Provision of health care for the rural populatich for tree of charge

☐ Focus on the development of curative ang ren aliitatite health care

☒ Promoting the participation of the private hector ii service delivery

Explanation :

The policy emphasizes decentralization, expanding health services to rural areas, and encouraging private sector involvement to complement government efforts in delivering healthcare services. Promoting the participation of the private sector ensures resource mobilization and improves access and quality of healthcare.

The focus of the 1993 policy was on primary healthcare, especially for rural populations, rather than specialized tertiary care hospitals. The policy didn't promise free healthcare but emphasized accessibility and equity. The policy shifted emphasis towards preventive and promotive care, aligning with global primary healthcare trends.

- Transitional Government of Ethiopia (1993): "Health Sector Development Policy."

A 72-year-old man came to a clinic with a swelling under his chin associated with fever and swallowing difficulty of three days duration. Physical examination showed a 3cm x 4 cm firm, tender and diffuse submandibular mass associated with limited capability of opening his mouth. What is the most likely microbe responsible for this patient's problem?

☒ Polymicrobial

☐ Aerobic



A 72-year-old man came to a clinic with a swelling under his chin associated with fever and swallowing difficulty of three days duration. Physical examination showed a 3cm x 4 cm firm, tender and diffuse submandibular mass associated with limited capability of opening his mouth. What is the most likely microbe responsible for this patient's problem?

☒ Polymicrobial

☐ Aerobic

☐ Gram-negatives

☐ Anaerobes

Explanation :

This presentation suggests Ludwig's angina, a severe cellulitis of the submandibular and sublingual spaces. Ludwig's angina is almost always polymicrobial, involving a mix of anaerobic and aerobic bacteria from the oral flora. Common organisms include Streptococcus, Staphylococcus, and Bacteroides species. Immediate broad-spectrum antibiotics covering aerobic and anaerobic organisms (e.g., ampicillin-sulbactam or clindamycin) and potential surgical drainage if there is airway compromise is the route of management.

- UpToDate: "Deep neck space infections, including Ludwig's angina."

A patient came to a hospital with four lower anterior teeth displaced lingually along with their socket. He gave a history of a fall down accident. However, radiograph investigation showed no root fracture. What is the most likely diagnosis to explain the finding?

☐ Alveolar bone fracture

☐ Avulsion

☒ Subluxation

☐ Ligament disruption

Explanation :

Subluxation is the loosening of a tooth due to injury, resulting in damage to the periodontal ligament but no displacement from the alveolus or socket fracture. The teeth in this case are displaced but remain within the socket, and no root fracture is noted, fitting the description of subluxation. Displacement lingually without root fracture on radiographs indicates the periodontal ligament has been disrupted. This is consistent with subluxation, not avulsion or alveolar bone fracture. Immediate repositioning of displaced teeth is important. Splinting may be required for stabilization, along with follow-up to assess pulp vitality.

- UpToDate, Andreasen JO. Traumatic Dental Injuries: A Manual.

A 25-year-old male patient came to a medical OPD with complaints of shortness of breath, palpitation and generalized body swelling for two weeks. Evaluation result show, temperature 36.4°C, respiratory rate of 28/min, pulse rate of 98/min, and blood pressure of 100/60mmHg. There is a diastolic murmur at the apex. Echocardiography shows LVEF of 35 percent, MVA of 1.5 cm², mitral valve leaflets are shortened and thickened. What is the most likely underlying cause of this patient's illness?

☐ Atherosclerosis

☒ Rheumatic heart disease



A 25-year-old male patient came to a medical OPD with complaints of shortness of breath, palpitation and generalized body swelling for two weeks. Evaluation result show, temperature 36.4°C, respiratory rate of 28/min, pulse rate of 98/min, and blood pressure of 100/60mmHg. There is a diastolic murmur at the apex. Echocardiography shows LVEF of 35 percent, MVA of 1.5 cm², mitral valve leaflets are shortened and thickened. What is the most likely underlying cause of this patient's illness?

- ☐ Atherosclerosis
- ☒ Rheumatic heart disease
- ☐ Carcinoid tumor
- ☐ Dilated cardiomyopathy

Explanation :

The narrowed mitral valve area (1.5 cm²) with thickened and shortened leaflets strongly suggests mitral stenosis, which is a hallmark of chronic rheumatic heart disease (RHD). Left ventricular ejection fraction (LVEF) of 35% may reflect secondary heart failure due to mitral stenosis. Rheumatic fever can lead to progressive inflammation and fibrosis of the mitral valve leaflets, resulting in stenosis and regurgitation. Chronic rheumatic damage explains the mitral valve abnormalities in this case. Medical therapy with diuretics, beta blockers, or anticoagulants may be required. Definitive treatment involves percutaneous balloon mitral valvotomy or valve replacement in severe cases.

- UpToDate: "Clinical manifestations and diagnosis of rheumatic heart disease."

A primary hospital hired a new director about six months ago. He is upset by the fact that immediately after leaving the main office in the morning, nurses congregate at a local coffee house. He called the head nurse to his office and made it clear that this "socializing during working hours" must end. The nurse told the director that she believes that the nurses' informal conversations over morning coffee are vital to her unit's high performance. The director at all convinced and he is set in his opinion because he does not want the nurse to win over him. The head nurse accepted his decision and informed her colleagues to stop gathering in the coffee house. What is the most likely conflict management style used by the new director?

- ☐ Collaboration
- ☐ Compromise
- ☒ Competition
- ☐ Avoidance

Explanation :

Competition refers to an assertive and uncooperative approach to conflict resolution where one party prioritizes their own goals over others. The director disregards the head nurse's perspective and insists on enforcing the decision to stop gatherings, highlighting a win-lose mindset. High concern for achieving personal objectives. Low concern for collaboration or compromise with others.

- UpToDate: "Conflict management in healthcare settings."

A community health team planned to give health education activities in a district. The team prioritizes different health problems based on whichever problems could be manageable with the



A community health team planned to give health education activities in a district. The team prioritizes different health problems based on whichever problems could be manageable with the available resources. What is the most likely criterion of problem prioritization used in the scenario?

- ☐ Community concern
- ☐ Severity
- ☐ Magnitude
- ☒ Feasibility

Explanation :

Feasibility refers to the practicality of addressing a problem given the available resources, including finances, personnel, materials, and infrastructure. In this scenario, the team focuses on managing problems that can realistically be addressed with their current resources, making feasibility the most relevant criterion. Feasibility focusses on resource availability and constraints. Ensures that planned interventions are achievable, maximizing the use of limited resources.

- UpToDate: "Approaches to prioritizing community health interventions."

During supervision, a director of a new primary hospital identified that nurses working in an obstetrics and gynecology Ward usually did not use partograph during labor and delivery due to lack of adequate knowledge and skill to apply the instrument. The director decided to organize a five day on job training for all nurses in the hospital on how to use the instrument. At the end, a post assessment by the trainers on the knowledge and skill of using the instrument indicated that the training was effective. What is the most likely component of monitoring and evaluation assessed by the trainers in this case?

- ☐ Output
- ☒ Outcome
- ☐ Impact
- ☐ Process

Explanation :

Outcome refers to the short- to medium-term effects of an intervention. In this case, the improvement in knowledge and skills among the nurses after the training reflects the direct result of the educational intervention.

A 60-year-old male patient, who is on his third postoperative day for colonic cancer, has developed abdominal pain and vomiting. The pain is at the epigastric area and it radiates to the back. Abdominal ultrasound shows thickened gall bladder wall and peri-cholecystic fluid collection but no gallstone. CBC shows WBC count of 19000cells/mm³. What is the most effective management for this patient?

- ☐ Exploratory laparotomy and colostomy
- ☒ Fluid resuscitation and IV antibiotic
- ☐ IV antibiotics and Immediate cholecystectomy
- ☐ Exploratory laparotomy

74

A 60-year-old male patient, who is on his third postoperative day for colonic cancer, has developed abdominal pain and vomiting. The pain is at the epigastric area and it radiates to the back. Abdominal ultrasound shows thickened gall bladder wall and peri-cholecystic fluid collection but no gallstone. CBC shows WBC count of 19000cells/mm³. What is the most effective management for this patient?

- ☐ Exploratory laparotomy and colostomy
- ☒ Fluid resuscitation and IV antibiotic
- ☐ IV antibiotics and immediate cholecystectomy
- ☐ Exploratory laparotomy

Explanation :

The thickened gallbladder wall, peri-cholecystic fluid collection, and elevated WBC count in the absence of gallstones are classic findings of acalculous cholecystitis. This condition is more common in critically ill or postoperative patients. Initial step of management is fluid resuscitation and IV antibiotics are the most effective initial treatments. These measures address the systemic inflammation and infection. Broad-spectrum antibiotics covering gram-negative and anaerobic organisms (e.g., piperacillin-tazobactam or ceftriaxone with metronidazole) should be initiated. For patients who do not improve or deteriorate, percutaneous cholecystostomy (drainage) is the next step definitive management. Cholecystectomy may be considered in stable patients after resolution of acute inflammation.

- UpToDate: "Acalculous cholecystitis: Diagnosis and management."

75

A young girl in a waiting room of a health center is spontaneously acting-out as a pregnant woman to feel the experience of being pregnant and teach others. What is the most likely method of education used in this scenario?

- ☐ Demonstration
- ☐ Fablea
- ☒ Role-play
- ☐ Drama

Explanation :

Role-play is an educational method where participants act out scenarios or roles to understand or convey an experience, concept, or process. In this case, the girl is spontaneously acting as a pregnant woman to simulate the experience and teach others. Role-play encourages active participation, facilitates experiential learning by immersing participants in the role being taught, develops empathy and understanding of the perspective of others.

76

A 27-year-old female patient came to a medical OPD with complaints of reddish discoloration of skin, which is exacerbated by exposure to sun light and arthralgia of two years duration. On examination, there is a slightly raised erythematous rash on her cheeks, chest and arms. Her hand joints are tender and swollen. A laboratory result shows Hgb level of 8.1g/dl, WBC count of 4.1X10,000/micro litter and platelet count of 126,000/micro litter. What is the most specific antibody associated with this disease?

- ☐ Rheumatoid factor



76

A 27-year-old female patient came to a medical OPD with complaints of reddish discoloration of skin, which is exacerbated by exposure to sun light and arthralgia of two years duration. On examination, there is a slightly raised erythematous rash on her cheeks, chest and arms. Her hand joints are tender and swollen. A laboratory result shows Hgb level of 8.1 g/dl, WBC count of $4.1 \times 10,000/\text{micro litter}$ and platelet count of $126,000/\text{micro litter}$. What is the most specific antibody associated with this disease?

☐ Rheumatoid factor

☒ Anti-dsDNA

☐ Anti-Ro

☐ Anti-histone

Explanation :

The clinical findings of photosensitivity, malar rash, and arthralgia along with anemia, leukopenia, and thrombocytopenia are strongly suggestive of SLE. SLE is a chronic autoimmune disease characterized by multi-organ involvement and production of autoantibodies. Anti-dsDNA is highly specific for SLE and correlates with disease activity, particularly lupus nephritis. Other autoantibodies may be present but lack the specificity of anti-dsDNA.

- UpToDate: "Clinical manifestations and diagnosis of systemic lupus erythematosus in adults."

77

A 25-year-old male patient came to a medical OPD with complaints of shortness of breath, palpitation and generalized body swelling for two weeks. Evaluation result show, temperature 36.4-degree centigrade, respiratory rate of 28/min, pulse rate of 98/min, and blood pressure of 100/60mmHg. There is a diastolic murmur at the apex. Echocardiography shows LVEF of 35 percent, MVA of 1.5 cm², mitral valve leaflets are shortened and thickened. What is the most likely underlying cause of this patient's illness?

☐ Atherosclerosis

☒ Rheumatic heart disease

☐ Carcinoid tumor

☐ Dilated cardiomyopathy

Explanation :

The findings of a diastolic murmur at the apex, mitral valve thickening, reduced MVA (indicative of mitral stenosis), and reduced left ventricular ejection fraction (LVEF) suggest rheumatic heart disease (RHD) as the underlying cause. RHD is a chronic complication of rheumatic fever, typically affecting the mitral valve. RHD commonly affects younger patients in endemic regions. Mitral stenosis is a hallmark, often accompanied by mitral regurgitation or other valvular lesions. Progressive valve damage leads to heart failure and atrial fibrillation.

- UpToDate: "Clinical manifestations and diagnosis of rheumatic heart disease."

78

A 57-year-old male patient came to a regular OPD with complaints of constipation and rectal bleeding. He is chronically sick looking and has pale conjunctiva. On per-rectal examination there is no fissure or hemorrhoid. What is the most appropriate diagnostic method in this patient's care?

☐ Abdomino-pelvic US

☒ Colonoscopy

Air contrast enema



A 57-year-old male patient came to a regular OPD with complaints of constipation and rectal bleeding. He is chronically sick looking and has pale conjunctiva. On per-rectal examination there is no fissure or hemorrhoid. What is the most appropriate diagnostic method in this patient's care?

- ☐ Abdomino-pelvic US
- ☒ Colonoscopy
- ☐ Air-contrast enema
- ☐ Fecal Occult blood test

Explanation :

Chronic constipation, rectal bleeding, and pallor suggest colorectal cancer or another lower gastrointestinal (GI) pathology, such as polyps. Pallor indicates anemia, likely due to chronic blood loss, further raising suspicion of malignancy. Colonoscopy is gold Standard for detecting and diagnosing colorectal cancer, as it provides direct visualization of the entire colon and rectum and allows for biopsies of suspicious lesions.

- UpToDate: "Overview of colonoscopy in colorectal cancer screening."

A 32-year-old male patient sustained road traffic accident after a car he was in rolled over many times. He was immediately brought to an emergency room and he was conscious and complaining right side flank pain and reddish discoloration of urine. What best investigation modality should be used to identify the cause of hematuria?

- ☒ CT with IV contrast
- ☐ Cystourethrogram
- ☐ Abdominopelvic ultrasound
- ☐ Plain abdominal X-ray

Explanation :

Hematuria and flank pain following trauma are suggestive of renal or urinary tract injury. Contrast-enhanced CT of the abdomen and pelvis is the gold standard for assessing renal, ureteral, and bladder injuries. Provides detailed imaging of renal parenchyma, vascular injuries, and urinary tract structures. Detects associated intra-abdominal injuries that may accompany blunt trauma.

- UpToDate: "Imaging of renal and genitourinary trauma."

In an area a significant number of pregnant mothers and children are found to be affected by micro-nutrient deficiency. As a long-term solution, people in the catchment area are encouraged to grow and consume various items of foods. What is most appropriate intervention applied in this scenario?

- ☐ Fortification
- ☒ Diversification
- ☐ Supplementation
- ☐ Modification

Explanation :

Refers to encouraging the consumption of a variety of locally available foods to improve dietary quality and address micronutrient deficiencies sustainably. Focuses on long-term changes in



In an area a significant number of pregnant mothers and children are found to be affected by micro-nutrient deficiency. As a long-term solution, people in the catchment area are encouraged to grow and consume various items of foods. What is most appropriate intervention applied in this scenario?

☐ Fortification

☒ Diversification

☐ Supplementation

☐ Modification

Explanation :

Refers to encouraging the consumption of a variety of locally available foods to improve dietary quality and address micronutrient deficiencies sustainably. Focuses on long-term changes in dietary habits and food production within the community.

Fortification involves adding micronutrients to commonly consumed foods (e.g., iodized salt, fortified flour), but it is not directly about growing various foods. Supplementation refers to providing specific vitamins or minerals (e.g., iron tablets, vitamin A capsules) and is typically a short-term solution. Modification involves altering food processing or preparation methods but does not encompass the broader goal of increasing dietary diversity.

- UpToDate: "Strategies to address micronutrient deficiencies."

A medical doctor planned to test the claim that the mean fasting blood sugar level of type II diabetic patients is 126 mg/dL. The doctor selected 36 diabetic patients randomly and their mean blood sugar level was found to 136 mg/dL with standard deviation of 9 mg/dL. The 95 percent CI for the population mean blood level was 138.94. What is the most appropriate conclusion based on the given evidences?

☐ The population mean blood sugar level of diabetic patients was greater than 136 mg/dL

☒ The population mean blood sugar level of diabetic patients was not 126 mg/dL

☐ The population mean blood sugar level of diabetic patients was 136 mg/d

☐ The population mean blood sugar level of diabetic patients was 126 mg/d

Explanation :

The provided 95% Confidence Interval (CI) does not include 126 mg/dL, suggesting that the population mean is statistically significantly different from 126 mg/dL at the 5% significance level. A CI that excludes the null hypothesis value means we reject the null hypothesis. The population mean is not equal to 126 mg/dL but is statistically likely to fall within the range of the given CI.

A 65-year-old male patient came with abdominal pain of three days duration. Associated with this he had abdominal distention and failure to pass feces and flatus. His pulse rate was 120/min and respiratory rate was 32/min. Plain abdominal x-ray showed coffee bean appearance, with the convexity of the loop lying in the right upper quadrant. What is the most likely diagnosis?

☐ Large bowel obstruction secondary to cecal volvulus

☒ Large bowel obstruction secondary to sigmoid volvulus

☐ Large bowel obstruction secondary to rectal cancer



A 65-year-old male patient came with abdominal pain of three days duration. Associated with this he had abdominal distention and failure to pass feces and flatus. His pulse rate was 120/min and respiratory rate was 32/min. Plain abdominal x-ray showed coffee bean appearance, with the convexity of the loop lying in the right upper quadrant. What is the most likely diagnosis?

- ☐ Large bowel obstruction secondary to cecal volvulus
- ☒ Large bowel obstruction secondary to sigmoid volvulus
- ☐ Large bowel obstruction secondary to rectal cancer
- ☐ Small bowel obstruction secondary to small bowel volvulus

Explanation :

Coffee bean appearance is a hallmark radiographic sign of sigmoid volvulus. The two limbs of the dilated sigmoid colon create the characteristic shape. Common in older patients (>60 years). Risk factors include chronic constipation, long-standing colonic distension, and neuropsychiatric conditions.

Cecal Volvulus also presents with a coffee bean sign but is less common. Typically affects younger patients (30–60 years). The convexity of the loop lies in the right upper quadrant, as described in the question.

A 24-year-old man, who was a known chronic liver disease patient taking Furosemide 80mg/d and Spironolactone 200mg/d, came to a follow up clinic with complaints of sleep disturbance and irritability of two weeks duration with recent altered mental status. On physical examination, he had confusion, ascites and asterixis. What is the most appropriate first step in the management of this patient?

- ☐ Perform therapeutic ascitic tap
- ☐ Empirically start on antibiotics

☒ Start on Lactulose

- ☐ Continue diuretics

Explanation :

A ten-year-old boy came to an OPD with cough, fever, breathing difficulty and loss of appetite. He was given antibiotics and anti-pyretic. At home, he became agitated, crying and uncomfortable around family members. His mom became uncomfortable and decided to check on him thoroughly which she realized that while being treated in the clinic, there was inappropriate examination on his private parts. She reported to the medical director next day and complained of inappropriate sexual harassment in the clinic. What must the medical director do to avoid such a scenario for next time?

- ☐ Take administrative procedure to the attending officer though there is no proof for sexual harassment
- ☐ Deny that the clinic has never done this kind of sexual harassment
- ☐ Apologize to the mother for what has happened already and discuss with his staff.

☒ The clinic has to develop a protocol, guideline and manual of handling minors.

Explanation :



A ten-year-old boy came to an OPD with cough, fever, breathing difficulty and loss of appetite. He was given antibiotics and anti-pyretic. At home, he became agitated, crying and uncomfortable around family members. His mom became uncomfortable and decided to check on him thoroughly which she realized that while being treated in the clinic, there was inappropriate examination on his private parts. She reported to the medical director next day and complained of inappropriate sexual harassment in the clinic. What must the medical director do to avoid such a scenario for next time?

- ☐ Take administrative procedure to the attending officer though there is no proof for sexual harassment
- ☐ Deny that the clinic has never done this kind of sexual harassment
- ☐ Apologize to the mother for what has happened already and discuss with his staff.
- ☒ The clinic has to develop a protocol, guideline and manual of handling minors.

Explanation :

This case raises serious allegations of inappropriate conduct, which highlight the lack of a clear protocol for handling minors during clinical examinations. Without defined policies, such incidents may recur, undermining trust in healthcare. Developing protocols, guidelines, and manuals ensures protection of patients and adherence to ethical standards. Minimizes ambiguity regarding appropriate behavior during examinations, especially for vulnerable populations like minors.

In which of the following cases is the individual's right to autonomy violated?

- ☐ A Jehovah's witness patient with symptomatic anemia who refuses blood transfusion
- ☐ B. An athlete who seeks a second opinion
- ☒ C. An infant whose parents elect for surgery to correct a congenital heart defect

Explanation :

Autonomy refers to an individual's right to be self-governing. Although the procedures may be in the best interests of the patient, if the patient refuses and this refusal is informed, then the doctor or relatives cannot coerce that person into having the procedure.

A 28-year-old female patient came to a regular surgical OPD with a complaint of intermittent vomiting of six months duration. The vomiting was large in volume and of ingested mater. She had also marked weight loss and on and off type of epigastric pain for the last three years. Physical examination showed significant malnutrition succussion splash was positive. What is the most likely diagnosis for this patient?

- ☐ GOO secondary to gastric cancer
- ☐ GERD
- ☒ GOO secondary Chronic PUD
- ☐ Gastric Volvulus

Explanation :

GOO a condition where the gastric contents cannot pass through the pylorus into the duodenum. Chronic PUD the most common cause of GOO in this demographic, due to scarring and fibrosis from repeated ulcer healing. Intermittent vomiting of ingested food suggests mechanical



86

A 28-year-old female patient came to a regular surgical OPD with a complaint of intermittent vomiting of six months duration. The vomiting was large in volume and of ingested mater. She had also marked weight loss and on and off type of epigastric pain for the last three years. Physical examination showed significant malnutrition succussion splash was positive. What is the most likely diagnosis for this patient?

☐ GOO secondary to gastric cancer

☐ GERD

☒ GOO secondary Chronic PUD

☐ Gastric Volvulus

Explanation :

GOO a condition where the gastric contents cannot pass through the pylorus into the duodenum. Chronic PUD the most common cause of GOO in this demographic, due to scarring and fibrosis from repeated ulcer healing. Intermittent vomiting of ingested food suggests mechanical obstruction distal to the stomach. Weight loss and malnutrition reflect chronicity and nutritional impairment.

- UpToDate: "Gastric outlet obstruction in adults."

87

A hospital director and a head of a pharmacy department are discussing to determine the quantity of medical supplies to be ordered. They list out constant and variable factors that influence order quantity. What is the most likely factor that makes the order quantity unreliable?

☒ Supplier lead time

☐ Storage capacity

☐ Order frequency

☐ Disease outbreaks

Explanation :

Supplier lead time: Refers to the time between placing an order and receiving it. Delays make accurate inventory planning challenging and lead to stockouts or overstocking, which can directly impact healthcare delivery.

Disease outbreaks: Sudden increases in demand for specific supplies (e.g., antibiotics, IV fluids) during outbreaks make order quantity predictions unreliable. Epidemics and pandemics strain supply chains due to higher demand and supply-side disruptions.

- UpToDate: "Hospital inventory and supply chain management."

88

A 22-year-old female patient came to a surgical OPD with a complaint of groin swelling of three days duration. On evaluation, there was 2x3cm globular mass just below the inguinal ligament. The mass bulges out while coughing. What is the most likely diagnosis of this finding?

☐ Hydrocele of the cord

☐ Vaginal hydrocele

☐ Inguinal hernia

☒ Femoral hernia

Explanation :



A 22-year-old female patient came to a surgical OPD with a complaint of groin swelling of three days duration. On evaluation, there was 2x3cm globular mass just below the inguinal ligament. The mass bulges out while coughing. What is the most likely diagnosis of this finding?

- ☐ Hydrocele of the cord
- ☐ Vaginal hydrocele
- ☐ Inguinal hernia
- ☒ Femoral hernia

Explanation :

Femoral hernia is common in women due to the wider pelvis. Anatomical location is below the inguinal ligament, where femoral structures pass. Bulging on coughing supports a hernia diagnosis. Inguinal hernia:

Less likely in this case due to the location below the inguinal ligament, but it must be considered because clinical differentiation can sometimes be challenging.

- UpToDate: "Evaluation of groin masses."

A 25-year-old patient, who had sustained bullet injury to the abdomen, came to an emergency room lately. Physical examination revealed that he has blood pressure of 70/30mmHg, pulse rate of 130/min, respiratory rate of 30/min, and abdominal examination showed rigidity and diffused tenderness. Which organ is most likely injured in this case?

- ☒ Liver
- ☐ Spleen
- ☐ Small bowel
- ☐ Large bowel

Explanation :

Gunshot wounds to the abdomen are associated with damage to highly vascular organs. The liver is the largest intra-abdominal organ and is highly vascularized, making it the most commonly injured organ in penetrating trauma.

- UpToDate: "Abdominal trauma in adults."

A 65-year-old construction worker who had sustained fall down injury from stairs, was diagnosed to have seven consecutive fractures on chest CT. What is the most appropriate approach for this patient?

- ☒ Should be admitted to ICU
- ☐ Surgical fixation is mandatory
- ☐ Start deep breath exercise immediately
- ☐ Treat as pulmonary contusion

Explanation :

Flail chest is likely diagnosis with seven consecutive fractures, leading to paradoxical chest wall movement and potential respiratory failure. These patients are at high risk for complications, including pulmonary contusion, respiratory distress, and infection. ICU admission is essential for close monitoring of respiratory function, oxygenation, and potential need for mechanical ventilation.



A 65-year-old construction worker who had sustained fall down injury from stairs, was diagnosed to have seven consecutive fractures on chest CT. What is the most appropriate approach for this patient?

☒ Should be admitted to ICU

☐ Surgical fixation is mandatory

☐ Start deep breath exercise immediately

☐ Treat as pulmonary contusion

Explanation :

Flail chest is likely diagnosis with seven consecutive fractures, leading to paradoxical chest wall movement and potential respiratory failure. These patients are at high risk for complications, including pulmonary contusion, respiratory distress, and infection. ICU admission is essential for close monitoring of respiratory function, oxygenation, and potential need for mechanical ventilation.

- UpToDate: "Management of rib fractures and flail chest."

A 35-year-old female patient, who was on her 3rd postoperative day after near total thyroidectomy was done for the diagnosis of Graves's disease, has started to complain circumoral and fingertip numbness. Her physical examination reveals positive Trousseau's sign with carpopedal spasm. Laboratory result shows serum calcium level of 7 mg/dL. What is the most appropriate next step to treat this patient?

☐ Treat with Thyroxine

☐ Treat with Oral calcium

☒ Treat with IV Calcium gluconate

☐ Reassure the patient that it will resolve

Explanation :

Normal calcium: 8.5–10.2 mg/dL; this patient has severe hypocalcemia (<8 mg/dL). Post-thyroidectomy hypocalcemia is likely due to transient hypoparathyroidism, a common complication of thyroid surgery. Symptoms like circumoral numbness, paresthesia, and Trousseau's sign are indicative of hypocalcemia. Optimal Management with IV calcium gluconate rapidly corrects symptomatic hypocalcemia. Once stabilized, oral calcium and vitamin D (calcitriol) should be initiated for long-term management.

- UpToDate: "Post-thyroidectomy hypocalcemia management."

A 27-year-old male patient who is a known CKD secondary to chronic glomerulonephritis is on follow up. His laboratory result shows hemoglobin of 4.7mg/dl and MCV of 75mg/dl. After further work up with Transferrin saturation and serum EPO level, the physician started him on iron with epoetin Alfa. What is the most appropriate target hemoglobin in this case?

☐ 9-10.5g/dl

☒ 10-11.5g/dl

☐ 7-8g/dl

☐ 8-9g/dl

A 27-year-old male patient who is a known CKD secondary to chronic glomerulonephritis is on follow up. His laboratory result shows hemoglobin of 4.7mg/dl and MCV of 75mg/dl. After further work up with Transferrin saturation and serum EPO level, the physician started him on Iron with epoiten Alfa. What is the most appropriate target hemoglobin in this case?

☐ 9-10.5g/dl

☒ 10-11.5g/dl

☐ 7-8g/dl

☐ 8-9g/dl

Explanation :

Caused by reduced erythropoietin production, compounded by iron deficiency and inflammation. Treat anemia to alleviate symptoms (e.g., fatigue, dyspnea) and avoid risks of overtreatment (e.g., cardiovascular complications). KDIGO Improving Global Outcomes) guidelines recommend maintaining hemoglobin between 10-11.5 g/dL to optimize outcomes. Levels >11.5 g/dL are associated with increased risk of thrombosis and hypertension.

- UpToDate: "Anemia management in chronic kidney disease."

A 50-year-old male patient, who had been smoking for the last 30 years, has come to a clinic with productive cough of blood tingled sputum since the last eight months. He has right side chest pain though he has no other complaint. On examination, there is wheeze over right mid lung field. PA and lateral CXR show heterogeneous opacification at middle lobe of right lung. What is the most appropriate next step to diagnosis the case?

☒ Chest CT-scan

☐ Sputum gene exper

☐ Bronchoscopy

☐ Chest MRI

Explanation :

The patient's age, long smoking history, and symptoms (hemoptysis, chest pain) are highly suspicious for malignancy. CXR abnormalities (heterogeneous opacification) warrant further characterization. Chest CT should be the next step. It is superior to CXR for assessing the size, location, and invasiveness of lung lesions. Provides critical staging information (e.g., lymphadenopathy, pleural involvement).

- UpToDate: "Evaluation of patients with suspected lung cancer."

A 34-year-old male patient came to an OPD with complaints of diarrhea for the last six months for which he was treated with antibiotics but with no improvement. In the last month, he became irritable and developed rash around the mouth and perineum. On physical examination he has muscle wasting, and vesicle and pustular crusting with scaling around oral and anal area. What is the most likely deficient nutrient in this patient?

☐ Vitamin C

☐ Niacin

☒ Zinc

A 34-year-old male patient came to an OPD with complaints of diarrhea for the last six months for which he was treated with antibiotics but with no improvement. In the last month, he became irritable and developed rash around the mouth and perineum. On physical examination he has muscle wasting, and vesicle and pustular crusting with scaling around oral and anal area. What is the most likely deficient nutrient in this patient?

☐ Vitamin C

☐ Niacin

☒ Zinc

☐ Selenium

Explanation :

The clinical scenario is consistent with zinc deficiency. Key features supporting this diagnosis. Chronic diarrhea can lead to zinc loss, compounding the deficiency. Periorificial and perineal dermatitis is a hallmark of acrodermatitis enteropathica, a condition related to zinc deficiency. Zinc plays a role in protein metabolism and neurotransmitter function, and its deficiency leads to these systemic symptoms. Non-response to antibiotics suggests a nutritional cause rather than infection.

- UpToDate

A 52-year-old patient was admitted to a COVID ICU ward. He was intubated and is using Oxygen. The oxygen and the tube must be checked regularly to avoid any complication. The attending ICU nurse was very busy with other patients where she did not make a visit to this patient. Unfortunately, the intubation was blocked with secretion and consequently, the Oxygen level dropped and the patient died to that effect. How should the head of ICU respond to this situation?

☐ The head should inform the family about the cause of death and they will decide whether to take it to the legal process or not.

☒ The situation is very sensitive, this could happen when many patients are admitted in an ICU. Hence, the head can bring the case rather in "Death Audit" period for his team.

☐ The head should not disclose the incident; inform the family that the patient has died because of natural course of COVID.

☐ The head should take administrative decision to the nurse, tell the family the patient has died and later discuss the case for others to learn from it

Explanation :

Transparency: It is essential to disclose the facts to the family. Identifying the cause of the event (nurse being busy) can help improve future care delivery. The death audit allows the team to understand systemic gaps without targeting individuals, ensuring collective learning.

- UpToDate

A surgeon wants to prepare surgical site of a patient who is on a table. He has given spinal anesthesia for herniorrhaphy, and has planned to apply strict measures to prevent surgical site infection. What is the most appropriate initial step to be applied by the surgeon?

☐ Wear surgical glove

☐ Shave the hair on surgical field

☐ Wash the surgical field with Povidone iodine

A surgeon wants to prepare surgical site of a patient who is on a table. He has given spinal anesthesia for herniorrhaphy, and has planned to apply strict measures to prevent surgical site infection. What is the most appropriate initial step to be applied by the surgeon?

- ☐ Wear surgical glove
- ☐ Shave the hair on surgical field
- ☐ Wash the surgical field with Povidone iodine
- ☒ Wash his hands with medicated soap

Explanation :

Hand hygiene is the cornerstone of preventing surgical site infections (SSIs). It removes transient microorganisms and reduces the risk of contamination. This step is performed before any direct interaction with the patient or surgical field.

- Schwartz's 11th, UpToDate

A 34-year-old male patient came to a surgical OPD with a complaint of protruding mass per rectum. The mass reduced by itself and toilet paper was stained with blood. What is the most likely diagnosis?

- ☐ Fistula in ano
- ☒ Hemorrhoid
- ☐ Anal fissure
- ☐ Perianal abscess

Explanation :

The mass that reduces spontaneously is characteristic of second-degree internal hemorrhoids, which prolapse with straining but retract on their own. External hemorrhoids do not prolapse spontaneously as they are located distal to the dentate line. Bright red blood on toilet paper (painless bleeding) is typical of internal hemorrhoids, as the region lacks somatic innervation. The absence of severe pain and other symptoms (e.g., fever, swelling) rules out conditions like perianal abscess and anal fissures.

Hemorrhoids are vascular cushions composed of dilated veins of the rectal and anal venous plexuses. Internal hemorrhoids arise above the dentate line, while external hemorrhoids occur below it. Risk factors include straining during defecation, constipation, pregnancy, and increased intra-abdominal pressure.

Classifications of Hemorrhoids:

- First-degree: No prolapse, bleeding may occur.
- Second-degree: Prolapse with defecation, spontaneously reduces.
- Third-degree: Prolapse requires manual reduction.
- Fourth-degree: Prolapse is irreducible.

Management is topical creams containing hydrocortisone or anesthetics. Surgical approach includes rubber band ligation, sclerotherapy, or surgical hemorrhoidectomy (for advanced stages).

Fistulas typically present with a chronic discharge of pus or stool through an external opening near the anus, not a protruding rectal mass. Bleeding is rare, and there is no spontaneous reduction of a mass. Anal fissures cause severe pain during defecation and are often associated with minor bleeding. There is no protruding mass with fissures. Perianal abscess presents with severe localized pain, swelling, erythema, and sometimes fever. A protruding mass that reduces spontaneously is not characteristic.

- Schwartz's 11th, UpToDate

A 75-year-old male patient was brought to a surgical regular OPD with a complaint of epigastric pain that radiates to the back of two days duration. He had some improvement while he was leaning forward. Previously, he had abdominal pain and he was told to have stones on his gall bladder but had no follow up for this condition. On evaluation, he had mild epigastric tenderness. His random blood glucose level was 300 mg/dl and WBC count was 20,000cells/mm³. Serum amylase and lipase was elevated 8 times each. What is the best management for this patient?

- ☐ Admit to ICU for IV antibiotics then PO/NG tube feeding
- ☒ Adequate resuscitation then keep him NPO for at least for 3 days
- ☐ Keep him in the ward, IV antibiotics, then keep NPO for 3 days
- ☐ Admit to ICU, keep him NPO for three days and IV antibiotics

Explanation :

99

A 30-year-old male was brought to an emergency room after he sustained fall down accident from six-meter-high tree. He was in distress and physical examination revealed a blood pressure of 90/60 mmHg, pulse rate of 120/min, respiratory rate of 38/min, oxygen saturation of 80 percent and absent air entry over the entire right lung. Chest X ray showed the whole right-side haziness. What is the most likely diagnosis of this patient?

- ☐ Tension pneumothorax
- ☒ Hemothorax
- ☐ Lung Contusion
- ☐ Simple pneumothorax

Explanation :

Trauma history fall from a height is a high-risk mechanism for thoracic injuries. Hypotension (90/60 mmHg) and tachycardia (120/min): Indicate hemorrhagic shock, which is consistent with massive blood loss into the thoracic cavity. Absent air entry over the right lung suggests fluid or air accumulation compressing the lung. Chest X-ray finding strongly suggests fluid accumulation (blood in hemothorax), differentiating it from pneumothorax, which would show hyperlucency. Immediate chest tube placement to evacuate blood and restore lung function.
Bailey 26th, UpToDate,

100

A 19-year-old high school student with recent behavioral change has been brought to a medical OPD by his family. The student has unexplained daydreaming and a decline in school performance recognized by the teacher. Evaluation result show, temperature of 36.7-degree centigrade, respiratory rate of 20/min, pulse rate of 70/min, and blood pressure of 120/80mmhg. There is no other remarkable finding. Laboratory results show that: Hgb level 12.6g/dl, WBC count of 8.8x10,000/micro litter and platelet count of 310,000/micro litter. Other biochemical tests are normal. What is the most likely diagnosis of the student?

- ☒ Absence seizures
- ☐ Syncope
- ☐ Benign paroxysmal vertigo
- ☐ Transient ischemic attack



amylase and lipase was elevated 6 times each. What is the best management for this patient?

☐ Admit to ICU for IV antibiotics then PO/NG tube feeding

☒ Adequate resuscitation then keep him NPO for at least for 3 days

☐ Keep him in the ward, IV antibiotics, then keep NPO for 3 days

☐ Admit to ICU, keep him NPO for three days and IV antibiotics

Explanation :

99

A 30-year-old male was brought to an emergency room after he sustained fall down accident from six-meter-high tree. He was in distress and physical examination revealed a blood pressure of 90/60 mmHg, pulse rate of 120/min, respiratory rate of 38/min, oxygen saturation of 80 percent and absent air entry over the entire right lung. Chest X ray showed the whole right-side haziness. What is the most likely diagnosis of this patient?

☐ Tension pneumothorax

☒ Hemothorax

☐ Lung Contusion

☐ Simple pneumothorax

Explanation :

Trauma history fall from a height is a high-risk mechanism for thoracic injuries. Hypotension (90/60 mmHg) and tachycardia (120/min): Indicate hemorrhagic shock, which is consistent with massive blood loss into the thoracic cavity. Absent air entry over the right lung suggests fluid or air accumulation compressing the lung. Chest X-ray finding strongly suggests fluid accumulation (blood in hemothorax), differentiating it from pneumothorax, which would show hyperlucency. Immediate chest tube placement to evacuate blood and restore lung function.

Bailey 26th, UpToDate,

100

A 19-year-old high school student with recent behavioral change has been brought to a medical OPD by his family. The student has unexplained daydreaming and a decline in school performance recognized by the teacher. Evaluation result show, temperature of 36.7-degree centigrade, respiratory rate of 20/min, pulse rate of 70/min, and blood pressure of 120/80mmhg. There is no other remarkable finding. Laboratory results show that: Hgb level 12.6g/dl, WBC count of $8.8 \times 10,000/\text{micro litter}$ and platelet count of $310,000/\text{micro litter}$. Other biochemical tests are normal. What is the most likely diagnosis of the student?

☒ Absence seizures

☐ Syncope

☐ Benign paroxysmal vertigo

☐ Transient ischemic attack

Explanation :

This patient has unexplained daydreaming episodes and a decline in school performance, both of which are hallmark features of absence seizures. Daydreaming behavior reflects episodes of loss of consciousness during seizures. Decline in school performance indicates cognitive interference caused by frequent undiagnosed seizures. An EEG (electroencephalogram) is diagnostic, showing generalized 3 Hz spike-and-wave discharges during episodes.

- UpToDate



33

A 45-year-old man presents to the emergency department with a severe headache, nausea, and vomiting. He has a history of ulcerative colitis and is on chronic steroid therapy. On examination, he has papilledema, focal neurologic deficits, and signs of increased intracranial pressure. What is the best imaging modality to confirm the diagnosis of sagittal sinus thrombosis in this patient?

- ☐ MRI angiography
- ☒ MRI venography
- ☐ Brain CT with contrast
- ☐ Brain CT without contrast

Explanation :

Sagittal sinus thrombosis is a rare condition that occurs when a blood clot forms in the superior sagittal sinus, the largest vein of the brain. It can cause symptoms of increased intracranial pressure, such as headache, nausea, vomiting, papilledema, and focal neurologic deficits. The diagnosis of sagittal sinus thrombosis is based on the imaging findings of a filling defect or absence of flow in the affected sinus. The best imaging modality to confirm the diagnosis is MRI venography, which is a technique that uses contrast-enhanced MRI to visualize the venous structures of the brain. MRI venography has a high sensitivity and specificity for detecting sagittal sinus thrombosis, and can also show the extent of the clot and the presence of collateral veins.

- MRI angiography is a technique that uses contrast-enhanced MRI to visualize the arterial structures of the brain. It is not useful for diagnosing sagittal sinus thrombosis, as it does not show the venous system.

- Brain CT with contrast is an alternative imaging modality for sagittal sinus thrombosis, but it is less sensitive and specific than MRI venography. It may show the empty delta sign, which is a triangular area of contrast enhancement in the posterior part of the sagittal sinus, surrounded by a hypodense thrombus.

- Brain CT without contrast is not reliable for diagnosing sagittal sinus thrombosis, as it may miss subtle findings or be confounded by artifacts. It may show the cord sign, which is a hyperdense sagittal sinus, or the dense vein sign, which is a hyperdense cortical vein draining into the sagittal sinus.

34

A 60-year-old woman with rheumatoid arthritis on NSAIDs for 2 years presents to the clinic with decreased urine output and elevated creatinine. She has no history of diabetes, hypertension, or kidney disease. She has normal blood pressure and no edema. Her urine analysis shows no protein, blood, or casts. What is the most likely mechanism of her renal impairment?

- ☐ Tubular necrosis
- ☒ Impaired renal autoregulation
- ☐ Interstitial nephritis
- ☐ Glomerulonephritis

Explanation :

NSAIDs are a class of drugs that inhibit the synthesis of prostaglandins, which are mediators of inflammation and pain. However, prostaglandins also play a role in maintaining renal blood flow and glomerular filtration rate, especially in conditions of low effective circulating volume, such as dehydration, heart failure, or liver cirrhosis. Prostaglandins dilate the afferent arterioles of the glomeruli, which increases the renal blood flow and glomerular filtration rate. NSAIDs block



34

A 60-year-old woman with rheumatoid arthritis on NSAIDs for 2 years presents to the clinic with decreased urine output and elevated creatinine. She has no history of diabetes, hypertension, or kidney disease. She has normal blood pressure and no edema. Her urine analysis shows no protein, blood, or casts. What is the most likely mechanism of her renal impairment?

- ☐ Tubular necrosis
- ☒ Impaired renal autoregulation
- ☐ Interstitial nephritis
- ☐ Glomerulonephritis

Explanation :

NSAIDs are a class of drugs that inhibit the synthesis of prostaglandins, which are mediators of inflammation and pain. However, prostaglandins also play a role in maintaining renal blood flow and glomerular filtration rate, especially in conditions of low effective circulating volume, such as dehydration, heart failure, or liver cirrhosis. Prostaglandins dilate the afferent arterioles of the glomeruli, which increases the renal blood flow and glomerular filtration rate. NSAIDs block this effect and impair the renal autoregulation, which leads to a decrease in renal blood flow and glomerular filtration rate. This can cause acute kidney injury or chronic kidney disease, especially in patients with underlying risk factors, such as older age, rheumatoid arthritis, or concomitant use of diuretics or ACEI.

- Tubular necrosis is a type of acute kidney injury that occurs when the tubular cells of the nephron are damaged by ischemia or toxins. It can be caused by NSAIDs, but it usually presents with oliguria, hematuria, proteinuria, and muddy brown casts in the urine.

- Interstitial nephritis is a type of acute or chronic kidney injury that occurs when the interstitial tissue of the kidney is inflamed by drugs, infections, or immune-mediated processes. It can be caused by NSAIDs, but it usually presents with fever, rash, eosinophilia, and sterile pyuria.

- Glomerulonephritis is a type of acute or chronic kidney injury that occurs when the glomeruli of the kidney are inflamed by immune complexes, complement activation, or antibodies. It is not caused by NSAIDs, but it can be associated with rheumatoid arthritis. It usually presents with hematuria, proteinuria, and red blood cell casts in the urine.

35

A 50-year-old man with a history of alcohol abuse and hepatitis C infection presents to the emergency department with hematemesis and melena. He has a distended abdomen with shifting dullness and a positive fluid wave. He has spider angiomas, palmar erythema, and gynecomastia. He is confused and disoriented. His blood tests show anemia, thrombocytopenia, coagulopathy, and elevated liver enzymes. What is the most likely cause of his portal hypertension?

- ☒ Cirrhosis
- ☐ Portal vein thrombosis
- ☐ Schistosomiasis
- ☐ Budd-Chiari syndrome

Explanation :

Portal hypertension is an increase in the pressure in the portal vein, which drains blood from the gastrointestinal tract, spleen, pancreas, and gallbladder to the liver. Portal hypertension can cause complications such as variceal bleeding, ascites, hepatic encephalopathy, and hepatorenal



35

A 50-year-old man with a history of alcohol abuse and hepatitis C infection presents to the emergency department with hematemesis and melena. He has a distended abdomen with shifting dullness and a positive fluid wave. He has spider angiomas, palmar erythema, and gynecomastia. He is confused and disoriented. His blood tests show anemia, thrombocytopenia, coagulopathy, and elevated liver enzymes. What is the most likely cause of his portal hypertension?

- ☒ Cirrhosis
- ☐ Portal vein thrombosis
- ☐ Schistosomiasis
- ☐ Budd-Chiari syndrome

Explanation :

Portal hypertension is an increase in the pressure in the portal vein, which drains blood from the gastrointestinal tract, spleen, pancreas, and gallbladder to the liver. Portal hypertension can cause complications such as variceal bleeding, ascites, hepatic encephalopathy, and hepatorenal syndrome. The most common cause of portal hypertension is cirrhosis, which is scarring and fibrosis of the liver due to chronic liver injury. Cirrhosis can be caused by various etiologies, such as alcohol abuse, hepatitis B or C infection, autoimmune hepatitis, hemochromatosis, or Wilson disease. Cirrhosis impairs the liver function and causes increased resistance to the portal blood flow. This patient has clinical and laboratory features of cirrhosis and its complications.

- Portal vein thrombosis is a clot in the portal vein, which can obstruct the portal blood flow and cause portal hypertension. It can be caused by hypercoagulable states, malignancy, infection, trauma, surgery, or cirrhosis. It can present with abdominal pain, fever, splenomegaly, and variceal bleeding.

- Schistosomiasis is a parasitic infection caused by *Schistosoma* species, which are transmitted by freshwater snails. It can affect various organs, such as the urinary tract, the intestines, the liver, and the spleen. It can cause portal hypertension by inducing granulomatous inflammation and fibrosis of the portal vein and its branches. It is endemic in some regions of Africa, Asia, and South America. It can present with hematuria, dysuria, abdominal pain, diarrhea, hepatosplenomegaly, and variceal bleeding.

- Budd-Chiari syndrome is an obstruction of the hepatic veins or the inferior vena cava, which drains blood from the liver to the heart. It can cause portal hypertension by increasing the back pressure in the portal vein. It can be caused by hypercoagulable states, malignancy, infection, trauma, surgery, or pregnancy. It can present with abdominal pain, ascites, hepatomegaly, and jaundice.

36

A 55-year-old man with a history of alcohol abuse and hepatitis C infection presents to the clinic with abdominal distension, leg swelling, and shortness of breath. He has spider angiomas, palmar erythema, and gynecomastia on physical examination. His blood tests show anemia, thrombocytopenia, coagulopathy, and elevated liver enzymes. An abdominal ultrasound shows cirrhosis, portal hypertension, and ascites. How would you like to manage his cirrhotic ascites?

- ☒ Dietary sodium restriction and oral diuretics
- ☐ Large-volume paracentesis and intravenous albumin
- ☐ Transjugular intrahepatic portosystemic shunt (TIPS)
- ☐ Liver transplantation